

Current Limitations:

1. 🚫 High Computational Cost

Your system trains:

- Multiple client models
- A global model
- Prototype extraction + contrastive learning

👉 This leads to:

- High GPU/CPU usage
- Slow training across rounds
- Memory overhead (especially with ResNet-based models)

2. 🌐 Communication Overhead in Federated Learning

Each round involves:

- Sending models to clients
- Receiving updated weights
- Aggregating models

👉 Problems:

- Large model size (e.g., ResNetUNet)
- Many communication rounds

3. ⚖️ Client Data Heterogeneity (Non-IID Data)

Each client may have:

- Different patient distributions
- Different infection severity
- Different imaging quality

👉 This causes:

- Model bias
- Poor global generalization

5. ❓ Uncertainty Estimation Limitations

You use **entropy-based uncertainty**, but:

👉 Problems:

- Overconfident wrong predictions are not handled
- Entropy alone is not always reliable

6. 🧩 Prototype Learning Challenges

Foreground/background prototypes:

- May not capture complex variations
- Sensitive to noise in features

👉 Result:

- Poor clustering in feature space
- Weak guidance for unlabeled data

7. 🔒 Privacy Not Fully Guaranteed

Although federated learning is used:

👉 Risks:

- Model updates can leak information
- Gradient inversion attacks possible

Improvements can be made:

1. ⚡ Model Optimization (Efficiency)

✓ Use Lightweight Models:

Replace heavy backbone (ResNet34) with:

- MobileNetV3
- EfficientNet

👉 Benefits:

- Faster training
- Lower memory usage
- Better for edge devices

2. 📡 Communication Efficiency

✓ Techniques:

- Model compression (quantization, pruning)
- Gradient sparsification
- Reduce communication rounds

👉 Advanced idea:

- Federated averaging with partial updates
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3. Better Handling of Non-IID Data

✓ Use Personalized Federated Learning:

- Each client keeps a **personalized layer**

✓ Methods:

- Federated Averaging improvements
- Clustered federated learning

👉 Benefit:

- Better adaptation to local data
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4. Improved Semi-Supervised Learning

✓ Add Pseudo-Labeling:

- Generate labels for unlabeled data
- Use confidence thresholding

✓ Consistency Regularization:

- Same image → similar prediction under augmentation

👉 Popular method:

- Mean Teacher Model
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5. Advanced Uncertainty Estimation

Replace entropy with:

- Monte Carlo Dropout
- Bayesian Neural Networks

👉 Benefits:

- More reliable uncertainty
 - Better filtering of noisy predictions
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6. Better Prototype Learning

✓ Improvements:

- Multi-scale prototypes
- Class-wise prototypes (not just fg/bg)

✓ Dynamic Prototype Updating:

- Update prototypes per round adaptively
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7. 🧠 Use Transformer-Based Models

Replace CNN with:

- Vision Transformer
- Swin Transformer

👉 Benefits:

- Better global context understanding
 - Improved segmentation accuracy
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8. 🔒 Stronger Privacy Mechanisms

✓ Add:

- Differential Privacy
- Secure aggregation

👉 Protect against:

- Data leakage
- Reconstruction attacks

High-Impact Research Directions

1: Trust-Aware Federated Aggregation (Highly Recommended)

💡 Problem:

Current aggregation uses:

- similarity
- agreement

But:

👉 It does not consider prediction reliability + uncertainty together

✅ Your Contribution:

Propose a **Trust Score** per client:

$$T_i = \alpha \cdot \text{Agreement} + \beta \cdot \text{Similarity} - \gamma \cdot \text{Uncertainty}$$

Then aggregate:

$$W_{global} = \sum T_i \cdot W_i$$

🎯 Why this is publishable:

- Novel aggregation strategy
- Combines 3 signals (agreement + prototype + uncertainty)
- Works well in non-IID + semi-supervised setup

📌 Add:

- Ablation study:
 - without uncertainty
 - without similarity
 - full model

2: Dynamic Prototype Alignment (VERY STRONG)

💡 Problem:

Your prototypes are:

- static per round
- weak for heterogeneous clients

✅ Your Contribution:

Introduce cross-client prototype alignment

Idea:

- Align feature spaces across clients using:
 - global prototype memory
 - contrastive alignment loss

New Loss:

$$L = L_{dice} + \lambda_1 L_{contrastive} + \lambda_2 L_{prototype-align}$$



Why strong:

- Tackles non-IID problem directly
- Very relevant in federated learning research

3: Uncertainty-Guided Pseudo Labeling

💡 Problem:

You use entropy, but don't act strongly on it

✅ Contribution:

- Generate pseudo-labels ONLY when:
 - entropy < threshold
- Use them as soft labels

Extend:

- Adaptive threshold per client

4: Dual-Model Consistency (Advanced)

Inspired by:

- Mean Teacher Model
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 **Idea:**

Each client maintains:

- student model
- teacher model (EMA of student)

5: Client Personalization Layer

 **Problem:**

Global model may not fit all clients

 **Contribution:**

- Add client-specific layers
 - Shared backbone + personalized head
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 **Why important:**

- Hot research topic in federated learning
- Solves non-IID issue

1. Abstract (Publish-Ready)

Title (example):

Trust-Aware Federated Semi-Supervised Learning for Medical Image Segmentation

Abstract:

Medical image segmentation often suffers from limited labeled data and strict data privacy constraints, making centralized training infeasible. Federated learning addresses privacy concerns by enabling distributed model training; however, it faces challenges such as data heterogeneity and unreliable client updates, especially in semi-supervised settings.

In this work, we propose a Trust-Aware Federated Semi-Supervised Segmentation (TA-FSS) framework that integrates uncertainty estimation, prototype learning, and model agreement to improve global model aggregation. Specifically, we introduce a novel trust score that combines prediction agreement, feature similarity, and uncertainty to dynamically weight client contributions during aggregation. Additionally, we incorporate prototype-based contrastive learning to enhance feature consistency across labeled and unlabeled clients.

Extensive experiments on medical image segmentation tasks demonstrate that the proposed approach significantly outperforms standard federated averaging and baseline semi-supervised methods, particularly under non-IID data distributions. The results highlight the effectiveness of trust-aware aggregation in improving robustness, generalization, and learning efficiency in federated environments.

2. Introduction

Medical image segmentation plays a critical role in computer-aided diagnosis, particularly in applications such as infection localization and disease progression analysis. Recent advances in deep learning have significantly improved segmentation performance; however, these methods typically require large amounts of annotated data. In practice, obtaining high-quality medical annotations is expensive and time-consuming, leading to limited labeled datasets.

Federated learning has emerged as a promising paradigm to address data privacy concerns by enabling collaborative model training across multiple clients without sharing raw data. Despite its advantages, federated learning suffers from several challenges, including data heterogeneity (non-IID distributions), communication constraints, and inconsistent client updates, which degrade global model performance.

To further reduce reliance on labeled data, semi-supervised learning techniques have been integrated into federated settings. However, combining federated and semi-supervised learning introduces additional complexity. In particular, unlabeled clients may produce unreliable predictions, and naive aggregation strategies such as simple averaging fail to account for varying client reliability.

To address these challenges, we propose a Trust-Aware Federated Semi-Supervised Segmentation (TA-FSS) framework. The key idea is to evaluate and weight each client's

contribution based on a trust score, which incorporates three critical factors:

- (i) agreement with the global model,
- (ii) similarity of learned feature representations via prototypes, and
- (iii) prediction uncertainty.

Furthermore, we enhance representation learning using prototype-based contrastive learning, which encourages consistent feature embeddings across clients and improves segmentation performance in low-label regimes.

The main contributions of this work are summarized as follows:

- We propose a trust-aware aggregation mechanism that dynamically weights client updates using agreement, similarity, and uncertainty.
- We introduce a prototype-guided contrastive learning strategy to improve feature alignment across heterogeneous clients.
- We design a federated semi-supervised segmentation framework tailored for medical imaging tasks.
- We conduct extensive experiments demonstrating improved performance over baseline methods, particularly under non-IID data settings.

3. Architecture Diagram



